



Effects of pin tip-clearance on the performance of an enhanced microchannel heat sink with oblique fins and phase change material slurry



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ABSTRACT

The performance of an enhanced microchannel heat sink with sectional oblique fins and with a coolant that contains Nano-Encapsulated Phase Change Material (NEPCM) particles is investigated using a 3D conjugated heat transfer model. Three volume fractions from $\xi = 0$ (pure water) to 0.3 are studied to investigate the effects of presence of NEPCM particles. Then the effects of introducing tip-clearance to the heat sink on thermal and hydrodynamic performance is examined at low Reynolds numbers (less than 215). Four values of tip-clearance to channel width ratio (t/W_c) are investigated ranging from a no gap ($t/W_c = 0$) to $t/W_c = 0.74$. On the bottom wall of the heat sink a range of uniform and fixed temperatures of 299.15 K to 340.15 K are applied. The cooling and hydrodynamic performance of the heat sink is studied using Nusselt and Euler numbers, respectively. It was observed that using of NEPCM slurry in contrast with pure water, enhanced the cooling performance of the heat sink but increased the Euler number. Furthermore, it was seen that the introduction of tip-clearance to the heat sink, if the clearance ratio is chosen properly, has the potential to enhance the cooling performance and reducing the Euler number simultaneously.

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1. Introduction

Necessity to miniaturize the advanced computational instruments to suit the modern needs and the high heat generation rate of these microprocessors on one side, and the direct relation between the efficiency of the heat dissipation systems and the computational performance of the processors on the other side, shows the importance of innovative ideas in design of micro cooling systems. In the past decade, researches on enhancing the heat rejection rate of conventional microchannel heat sinks have been carried out and it seems there is an optimum efficiency level for the conventional microchannels which is numerically and experimentally obtained. Therefore, the heat dissipation techniques' ability might have the potential of becoming an obstacle on the path of computational speed progress. In order to create new opportunities to enhance the cooling ability of the currently available heat sinks, researchers have investigated the cooling system configuration, working fluids and other relevant parameters to improve the heat sink cooling performance [1–5]. Wang and Peles [5]

categorized the enhancing techniques into two groups: passive and active. They experimentally investigated the effect of a combined passive/active heat transfer enhancement scheme on the thermal performance of a microchannel. Seyf et al. [6] studied the effect of utilizing NEPCM slurry in a micro tube heat sink and through performing a validated numerical analysis proved that adding NEPCM particles to the working fluid boosts the Nusselt number of the system. They compared the total entropy generation rate as a function of Reynolds number for a set of mass concentration levels of NEPCM particles in PAO coolant. Kuravi et al. [7] performed a numerical analysis on a microchannel heat sink and achieved an acceptable agreement between the numerical result and experimental results reported by Goel et al. [8]. The results presented by Kuravi et al. illustrated that adding NEPCM particles to the slurry enhances the cooling efficiency.

Sabbah et al. [9] numerically investigated the length of the thermal boundary layer on a surface when Micro Encapsulated Phase Change Material (MEPCM) particles are added to the pure coolant. They suggested that the presence of MEPCM particles in the slurry could increase the thermal entry length and it consequently boosted the heat dissipation rate. Lee et al. [10] concluded that by using oblique fins that disrupted the velocity profile development, one could improve the heat dissipation rate of a microchannel. They fabricated

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Nomenclature

C_p	specific heat (J/kg K)
dp	particle diameter (m)
$D_h = \frac{2W_c H}{(W_c + H)}$	hydraulic diameter (m)
Eu	Euler number
h	heat transfer coefficient ($W m^{-2} K^{-1}$)
H	channel height (m)
k	thermal conductivity ($W m^{-1} K^{-1}$)
Nu	Nusselt number
p	pressure (Pa)
Pe	Peclet number
q	heat flux ($W m^{-2}$)
Re	Reynolds number
t	tip clearance size (m)
T	temperature (K)
T_1	lower melting temperature (K)
T_2	higher melting temperature (K)
T_{Mr}	melting range ($T_2 - T_1$) (K)
u	velocity in x direction ($m s^{-1}$)
v	velocity in y direction ($m s^{-1}$)

w	velocity in z direction ($m s^{-1}$)
W_c	channel width (m)

Greek symbols

ξ	volume fraction of nanoparticles %
ξ_m	mass fraction of nanoparticles %
ρ	density ($kg m^{-3}$)
μ	viscosity (Pa s)
ϕ	viscous dissipation

Subscripts

c	channel
eff	effective
f	fluid
in	inlet
out	outlet
p	particle
s	solid
w	bottom wall

two microchannels based on the new design and analyzed the fluid flow. The effects of the oblique channels on boundary layer growth process and heat transfer behavior of the microchannel were studied and an increase of the Nusselt number to two times more than the base values was reported. They also investigated the effect of the oblique channels in more details [11]. Rayes et al. [12] fabricated a microchannel and designed an appropriate experimental setup so that five different tip clearance configurations could be tested. Effects of the tip clearance presence on the rejected heat flux and pressure drop along the channels were examined; they compared them as a function of pumping power. Their research indicated that the tip clearance distance is a dominant parameter in the heat transfer rate of the microchannel. Moores et al. [13] studied the effect of tip clearance on the thermal and hydrodynamic performances of a circular pin fin array. The results showed that the geometric parameters significantly affected the heat transfer and flow behaviors of the heat sink. They illustrated that tip clearance may increase or decrease the heat transfer rate at a fixed Reynolds number. Min et al. investigated a conventional microchannel heat sink with tip clearance and reported that the ratio of tip clearance size to channel width (t/W_c) is the parameter that should be considered in design of the heat sinks with tip clearance [14]. They could find an optimum value of this ratio for their problem definition. The effect of tip clearance was also studied in [15,16] and its effect as a function of different parameters such as pumping power, fluid average velocity, Reynolds number, etc. was investigated.

In this study, the effects of tip clearance size and NEPCM slurry coolant on the performance of an enhanced microchannel heat sink with oblique channels have been studied. The effect of the NEPCM slurry on enhancement of the heat dissipation rate of the heat sink in three different bottom wall temperature conditions have been examined and the effect of the presence of tip clearance on fluid flow and heat transfer rate of the heat sink is reported.

2. Mathematical modeling and governing equations

Fig. 1 shows the schematic of the problem definition. The temperature of the working fluid at the inlet surface is below the melting point of the NEPCM particles and therefore, it is expected that all of the PCM particles are solid. In this research, NEPCM slurry consists of Octadecane as the NEPCM particles and distilled water as the base fluid.

The encapsulated shell is strong enough to keep the melted PCM inside and does not let the PCM to mix with the fluid. The area between two hatched red lines in Fig. 1 (right) shows the computational domain. The boundaries of the computational domain are considered to be periodic. It means that the shape in the computational domain is repeated along with the boundary surfaces at the same shape, which is in the domain. The dimensions used in this study are consistent to the ones used in Ref. [10]. The periodic computational domain consists of three rectangular volumes including the central region in which the oblique fins and the heated walls are located, the inlet block to ensure the entered flow to the central region is developed, and the outlet block. The thermophysical properties of Octadecane, the used NEPCM particles, is given in Table 1.

The following assumptions are made to solve this problem:

1. The flow is laminar and incompressible [10,11].
2. Because the concentration is below 0.3, so the Newtonian fluid characteristics are expected [17].
3. Radiation effect is neglected.
4. The velocity of the suspended nanoparticles and the flow velocity has been considered equal and therefore, the particles are not making local different velocities [7].
5. A homogenous distribution of particles over the fluid are expected [7].
6. The effect of the shell material of the capsules is neglected.
7. The NEPCM particles melts in a range for temperature from T_1 to T_2 .
8. No depletion layer effect is considered because the ratio of channel size to particle size is sufficiently large [18,19].
9. The micro-convection caused by the particle-wall, particle-particle, and particle-fluid interactions is negligible. The effect of micro-convection phenomenon is considered in effective thermal conductivity definition.
10. There is no viscous dissipation.

The governing equations used to analyze the problem are continuity equation, momentum equations, and energy equation.

Continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

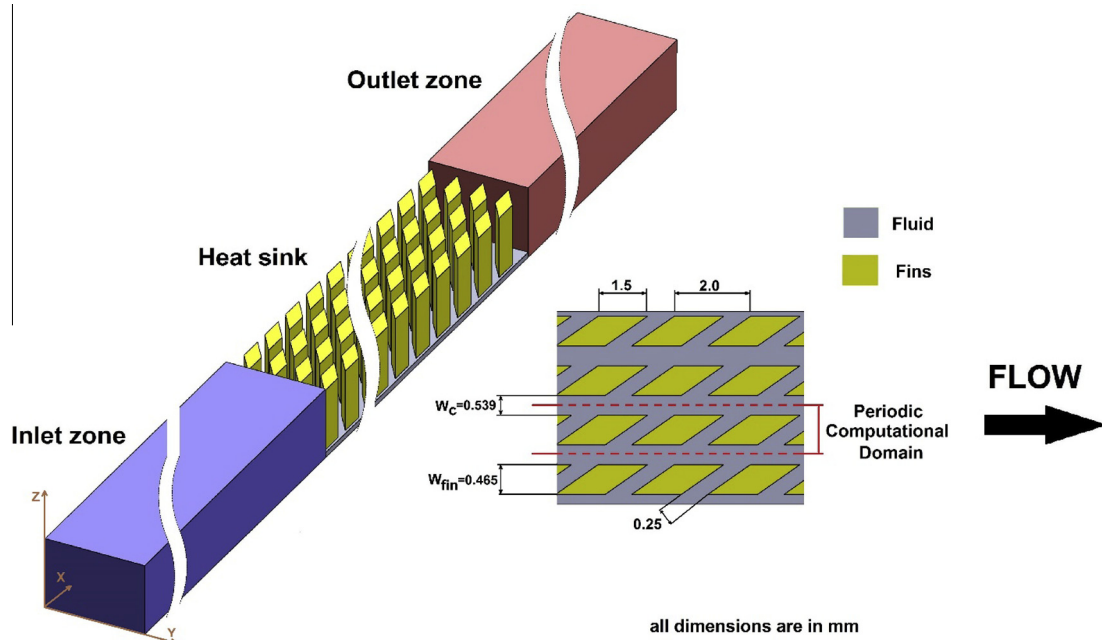


Fig. 1. Schematic of the physical layout of the heat sink (left) and a typical section view from heat sink area (right).

Table 1
Thermophysical properties of the NEPCM particles [6].

	Density (kg/m ³)	Specific heat J/kg K	Thermal conductivity W/m K	Latent heat J/kg
n-Octadecane	815	2000	0.18	244 × 10 ³

Momentum equations:

$$\rho_{eff} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left(\mu_{eff} \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu_{eff} \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu_{eff} \frac{\partial u}{\partial z} \right) \quad (2)$$

$$\rho_{eff} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x} \left(\mu_{eff} \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu_{eff} \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu_{eff} \frac{\partial v}{\partial z} \right) \quad (3)$$

$$\rho_{eff} \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \frac{\partial}{\partial x} \left(\mu_{eff} \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu_{eff} \frac{\partial w}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu_{eff} \frac{\partial w}{\partial z} \right) \quad (4)$$

Energy equation for the liquid region:

$$\rho_{eff} C_{p,eff} \left(u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} + w \frac{\partial T_f}{\partial z} \right) = \frac{\partial}{\partial x} \left(k_{eff} \frac{\partial T_f}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_{eff} \frac{\partial T_f}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_{eff} \frac{\partial T_f}{\partial z} \right) + \phi^2 \quad (5)$$

Energy equation for the solid region:

$$\frac{\partial}{\partial x} \left(k_s \frac{\partial T_s}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_s \frac{\partial T_s}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_s \frac{\partial T_s}{\partial z} \right) = 0 \quad (6)$$

The subscript “eff” denote the effective value of the thermo physical properties. The subscript “s” in Eq. (6) refers to the solid region. The ϕ^2 term in energy equation is viscous dissipation term and it represents the viscous energy dissipation rate per unit volume as listed below:

$$\phi^2 = \left(\left\{ 2 \left[\left(\mu_{eff} \frac{\partial u}{\partial x} \right)^2 + \left(\mu_{eff} \frac{\partial v}{\partial y} \right)^2 + \left(\mu_{eff} \frac{\partial w}{\partial z} \right)^2 \right] + \left(\mu_{eff} \frac{\partial u}{\partial y} + \mu_{eff} \frac{\partial v}{\partial x} \right)^2 + \left(\mu_{eff} \frac{\partial v}{\partial z} + \mu_{eff} \frac{\partial w}{\partial y} \right)^2 + \left(\mu_{eff} \frac{\partial w}{\partial x} + \mu_{eff} \frac{\partial u}{\partial z} \right)^2 \right\} \right)^{1/2} \quad (7)$$

The average heat transfer coefficient is expressed as:

$$\bar{h} = \frac{q}{A_{tot} \Delta T_{lm}} \quad (8)$$

where the log mean temp difference is expressed as:

$$\Delta T_{lm} = \frac{T_{out} - T_{in}}{\ln \left(\frac{T_w - T_{in}}{T_w - T_{out}} \right)} \quad (9)$$

In average heat transfer coefficient definition, q is the total thermal energy transferred to the coolant, which is passed through A_{tot} surface area that includes a portion of the bottom surface of the channel, which is not covered by fins, and the lateral surfaces area of fins.

Reynolds, Nusselt, and Euler numbers are determined from the relations:

$$Re = \frac{\rho_s u_{in} D_h}{\mu_s} \quad (10)$$

$$Nu = \frac{\bar{h} D_h}{K_s} \quad (11)$$

$$Eu = \frac{2 \Delta P}{\rho_s Re^2 N} \quad (12)$$

3. Slurry thermophysical properties

The thermophysical properties of distilled water as the base fluid are considered as temperature-dependent, but the properties of NEPCM particles, except heat capacity, are considered to be

constant. Temperature-dependent equations of distilled water, which are curve-fitted from well-established data [20] using least square technique, are as follows:

For $278.15 \text{ K} < T < 338 \text{ K}$

Density:

$$\rho = -0.002630252T^2 + 1.24115T + 861.6624 \quad (13)$$

Conductivity:

$$k = -9.455182 \times 10^{-6}T^2 + 0.007289006T - 0.7245947 \quad (14)$$

Heat capacity:

$$C_p = -8.732238 \times 10^{-5}T^3 + 0.09657741T^2 - 34.71345T + 8255.769 \quad (15)$$

Viscosity:

$$\mu = 1.548 \times 10^{-11}T^4 - 2.186052 \times 10^{-8}T^3 + 1.162394 \times 10^{-5}T^2 - 0.002763061T + 0.2484497 \quad (16)$$

The thermophysical properties of the slurry are defined as functions of thermophysical properties of the NEPCM particles and also that of the base fluid.

Density [1]:

$$\rho_{eff} = (1 - \xi_m)\rho_{water} + \xi_m\rho_p \quad (17)$$

Static thermal conductivity is calculated using Maxwell's relation [21]:

$$k_b = k_{water} \frac{2 + \frac{k_p}{k_{water}} + 2\xi \left(\frac{k_p}{k_{water}} - 1 \right)}{2 + \frac{k_p}{k_{water}} - \xi \left(\frac{k_p}{k_{water}} - 1 \right)} \quad (18)$$

$$\frac{k_{eff}}{k_b} = 1 + B\xi Pe_p^m$$

$$B = 3, \quad m = 1.5, \quad Pe_p < 0.67$$

$$B = 1.8, \quad m = 0.18, \quad 0.67 < Pe_p < 250 \quad (19)$$

$$B = 3, \quad m = \frac{1}{11}, \quad Pe_p > 250$$

where subscript w refers to the base fluid, water and p refers to the particles. Pe_p is particle Peclet number, which is defined as:

$$Pe_p = \frac{ed_p^2}{\alpha_{water}} \quad (20)$$

where α_{water} is thermal diffusivity of water and d_p is the particles' average diameter. e is the shear rate which is a function of all of the spatial coordinates and their relevant velocities. The shear rate magnitude is expressed as [22]:

$$e = \left[\frac{1}{2} \sum_i \sum_j \gamma_{ij} \gamma_{ji} \right]^{1/2} \quad (21)$$

where γ represents the shear rate. The above correlation for thermal conductivity of NEPCM slurries shows that the thermal conductivity of the slurry is defined based on two parameters: the shear rate magnitude and the particle size. Shear rate magnitude can be increased by increasing the fluid flow velocity or decreasing the conduction dimension. Particle size presents the interaction between the fluid molecules and particles including drag force, virtual mass, and lift force.

Heat capacity [6]:

$$C_{p,eff} = (1 - \xi_m)C_{p,water} + \xi_m C_{p,p} \quad (22)$$

Viscosity [23,24]:

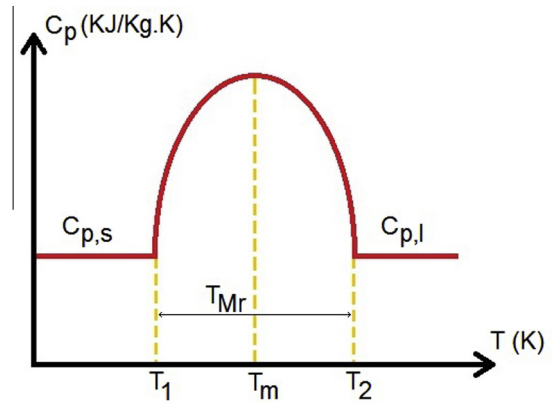


Fig. 2. The used sine profile to predict the specific heat of NEPCM slurry.

$$\mu_{eff} = \mu_w (1 - \xi - 1.16\xi^2)^{-2.5} \quad (23)$$

when NEPCM particles' temperature reaches T_1 – the start point of the melting region – the heat capacity of that in solid state increases based on the temperature, and after the heat capacity reached its maximum, then heat capacity reduces to the heat capacity of liquid NEPCM particles when it reaches T_2 , at the end of this region. Ali-setti and Roy [25] studied the effect of using different type of heat capacity profiles between the solidus and liquidus point and proved that it may vary the specific heat value of the slurry less than 4% and therefore, using any of the proposed profiles are acceptable. In present study, the authors used a sine profile to model the specific heat of NEPCM particles. The used profile in this study is shown in Fig. 2 and the sine equation for the specific heat of slurry is:

$$C_{p,p} = C_{p,pcm} + \left\{ \frac{\pi}{2} \cdot \left(\frac{h_{sf}}{T_{Mr}} - C_{p,pcm} \right) \cdot \sin \pi \left[\frac{(T - T_1)}{T_{Mr}} \right] \right\} \quad (24)$$

where $T_{Mr} = T_2 - T_1$ is the melting temperature range.

4. Boundary conditions

The inlet, outlet, and periodic boundary conditions are as below:

Inlet boundary condition:

$$u = U_{in}, \quad v = w = 0, \quad T_{in} = 296.15 \quad (25)$$

Outlet boundary condition:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x} = \frac{\partial w}{\partial x} = \frac{\partial T_f}{\partial x} = 0 \quad (26)$$

Periodic boundary condition:

$$u, v, \text{ and } w(\text{at } y = 0) = u, v, \text{ and } w(\text{at } y = W) \quad (27)$$

$$\begin{aligned} \frac{\partial u}{\partial y}(\text{at } y = 0) &= \frac{\partial u}{\partial y}(\text{at } y = W) \\ \frac{\partial v}{\partial y}(\text{at } y = 0) &= \frac{\partial v}{\partial y}(\text{at } y = W) \\ \frac{\partial w}{\partial y}(\text{at } y = 0) &= \frac{\partial w}{\partial y}(\text{at } y = W) \end{aligned} \quad (28)$$

Bottom wall temperature boundary condition:

$$T_f = T_s = T_{wall} \quad (29)$$

Walls boundary condition:

$$\text{Solid-liquid interfaces : } K_s \frac{\partial T_s}{\partial n} = K_f \frac{\partial T_f}{\partial n} \quad (30)$$

$$\text{Other walls : } K_s \frac{\partial T_s}{\partial n} = K_f \frac{\partial T_f}{\partial n} = 0 \quad (31)$$

5. Numerical solutions, grid independency and code validation

The used finite volume code employed for this research was previously validated and used in our research group and is explained in more detail in peer reviewed publications referenced at [6]. In this study, to discretize the three-dimensional governing equations, by utilizing finite volume method, conservation laws are applied over finite sized control volumes around grid points. Then, by integration over the volume, the relevant data are extracted. To discretize the convection and diffusion terms of energy and momentum equations, QUICK scheme was used. In this study, SIMPLE algorithm was employed based on an assumed pressure field, through using Navier–Stokes equations, the velocity distribution in three-dimensional space was calculated. Then, corrected values of pressure and velocity distribution are obtained from continuity equations. The convergence is defined based on the absolute errors of temperature, velocity components, and pressure values and the solution considered to have converged when these values become less than 10^{-8} , 10^{-7} , and 10^{-5} , respectively.

For inlet and outlet block, which include the coolant liquid and region walls, two structured grid are used and for the central region that contains the liquid and fins, an unstructured grid of tetrahedral volume elements was used. To ensure the independency of the results from grid number, four different grid sizes were examined: 662,105 (coarse), 1,022,911 (intermediate), 2,665,121 (fine) and 3,781,167 (very fine). As shown in Table 2, the difference between the obtained Nusselt and Euler numbers using fine grid and very fine grid is less than 1%; thus the fine grid is selected as a suitable grid size to conduct the simulations. The grid independency study is done for the case with bottom wall temperature of 299.15 K and velocity inlet of 0.250 m/s with pure water as coolant.

The results obtained from the developed finite volume (FV) code are compared and validated with the experimental data reported by Lee et al. [10], who did their experiment using an oblique finned microchannel with the specification shown in Fig. 1. They examined the thermal and fluid flow performance of the copper-based microchannel heat sink with a nominal main channel width of 0.539 μm and compared the obtained data with conventional microchannels with the same size and without oblique channels. The total transfer area is $25 \times 25 \text{ mm}^2$ on which a configuration of 39 rows of 20 oblique fins per row was fabricated. Utilized coolant was deionized water with a fixed inlet temperature of 20 °C. The bottom wall temperature is controlled using four cartridge heaters that their power input was monitored using a DC power supply unit. They used a polycarbonate cover with a thin silicone rubber layer to seal the top surface properly. The simulated portion in this study consists of one periodic part of the microchannel heat sink used in the experimental work of Lee et al. [10]. The fins and liquid around them are defined as solid and fluid regions, respectively. A constant static pressure equal to atmospheric pressure at the outlet, a uniform velocity and fixed temperature at the inlet, and constant temperatures similar to the ones used in the experiment at the bottom wall of the microchannel are defined.

Table 2
Grid independency study.

Number of grid	Euler number	Diff (%)	Nu	Diff (%)
662,105	2.4436212	–	12.2912984	–
1,022,911	2.1509624	13.61	10.9241971	12.51
2,665,121	1.95988523	9.75	10.0699926	8.48
3,781,167	1.9771022	0.87	10.1112568	0.41

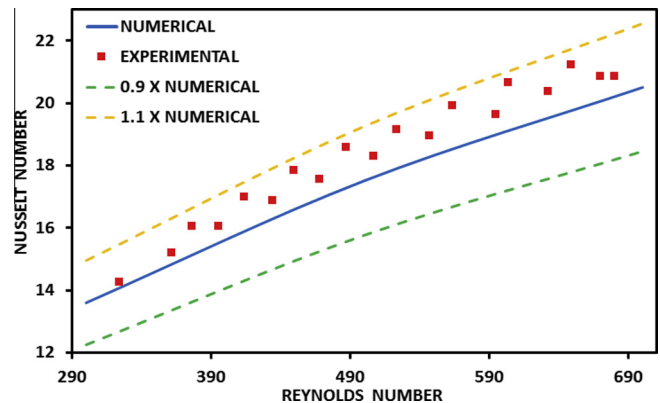


Fig. 3. The obtained results from FV code and the ones of experimental test.

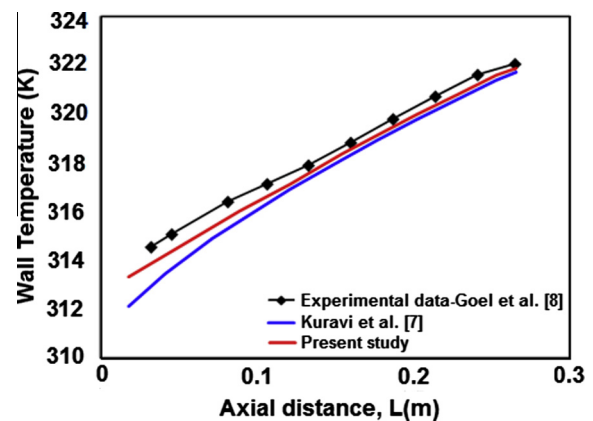


Fig. 4. Comparison of result using current model, numerical model [7] and experimental data [8].

The remaining variables are derived by extrapolating from the interior regions of the periodic computational domain. The acceptable consistency between the Nusselt number obtained from numerical simulation results and the ones of experimental data are illustrated and proved in Fig. 3. The dashed lines show $\pm 10\%$ deviation values compared to the numerical results. Considering the uncertainty values reported in the experimental report, the results obtained from the numerical code are in the acceptable range.

Fig. 4 shows the comparison among our results with experimental data of Goel et al. [8] and numerical calculation data reported by Kuravi et al. [7] for a PCM slurry through a pipe. The good consistency between the presented results in Fig. 4 proves the reliability of the used model.

6. Results and discussions

6.1. Effect of NEPCM slurry

We investigated how adding NEPCM particles to the pure water can change the cooling and hydrodynamics of the water first. Fig. 5 shows the variations of Nusselt number and bulk temperature versus Reynolds number at different bottom wall temperatures and Reynolds number at different NEPCM volume fractions.

It can be seen from Fig. 5 that when NEPCM particles are added to the base pure water coolant, the Nusselt number increases but bulk temperature decreases. The reason behind these opposite trends is that, when NEPCM particles are added to the working

liquid and when temperature of the particles reaches the liquid temperature, heat capacity of the particles and consequently slurry, as it is shown in Fig. 2, starts increasing. When NEPCM particles are added, two phenomena occur simultaneously to enhance the heat dissipation rate of the heat sink: (1) increase of heat capacity of the slurry that makes the coolant capable of absorbing more heat while it undergoes a lower temperature increase; and (2) since lower bulk-temperature makes a higher temperature gradient between the bottom hot wall and coolant bulk temperature, it boosts up the heat dissipation rate of the heat sink. Fig. 5 also shows that the NEPCM concentration has a direct relation with the system Nusselt number. When the velocity of the flow increases, the thickness of the generated thermal boundary layers decreases and it results in a tendency to increase the heat transfer

rate. In other words, for higher Reynolds numbers because of the thinner boundary layer on the surfaces, a higher cooling performance of the system is expected. So in terms of cooling performance of the heat sink, increase in NEPCM particles volume fraction leads to a higher cooling performance and lower bulk temperature. Regarding average bulk temperature of the coolant, when Reynolds number or NEPCM particles concentration increases, bulk temperature reduces accordingly.

Fig. 6 illustrates the variation of the Euler number with the NEPCM particles concentration at different inlet velocities. As expected, increasing the flow velocity -when the temperature of the wall is fixed- leads to lower fluid bulk temperature that consequently increases the viscosity of the slurry. This increase in viscosity yields a higher Euler number. Euler number as the

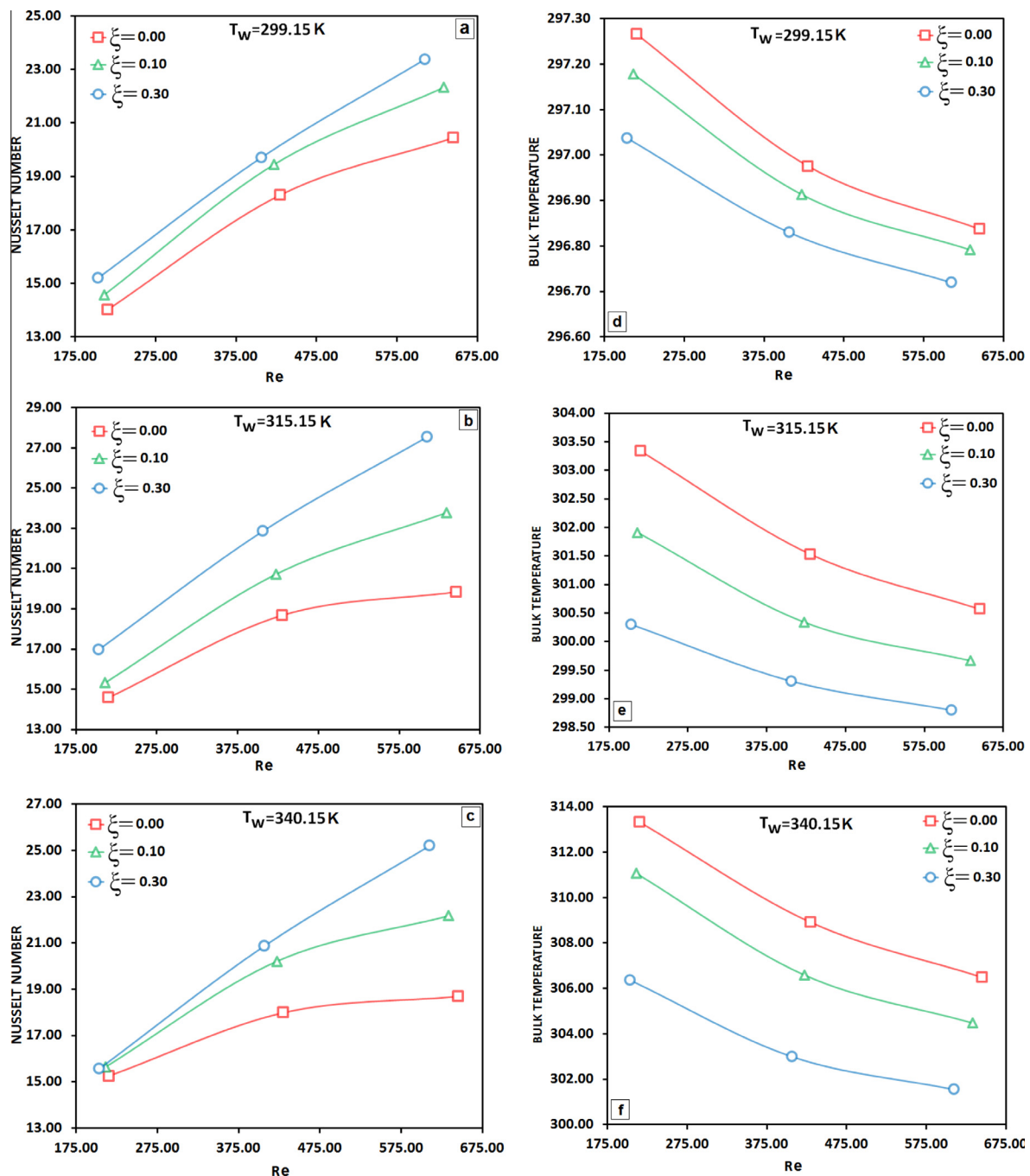


Fig. 5. Nusselt number and bulk temperature variations versus Reynolds number at different NEPCM volume fractions and bottom wall temperatures.

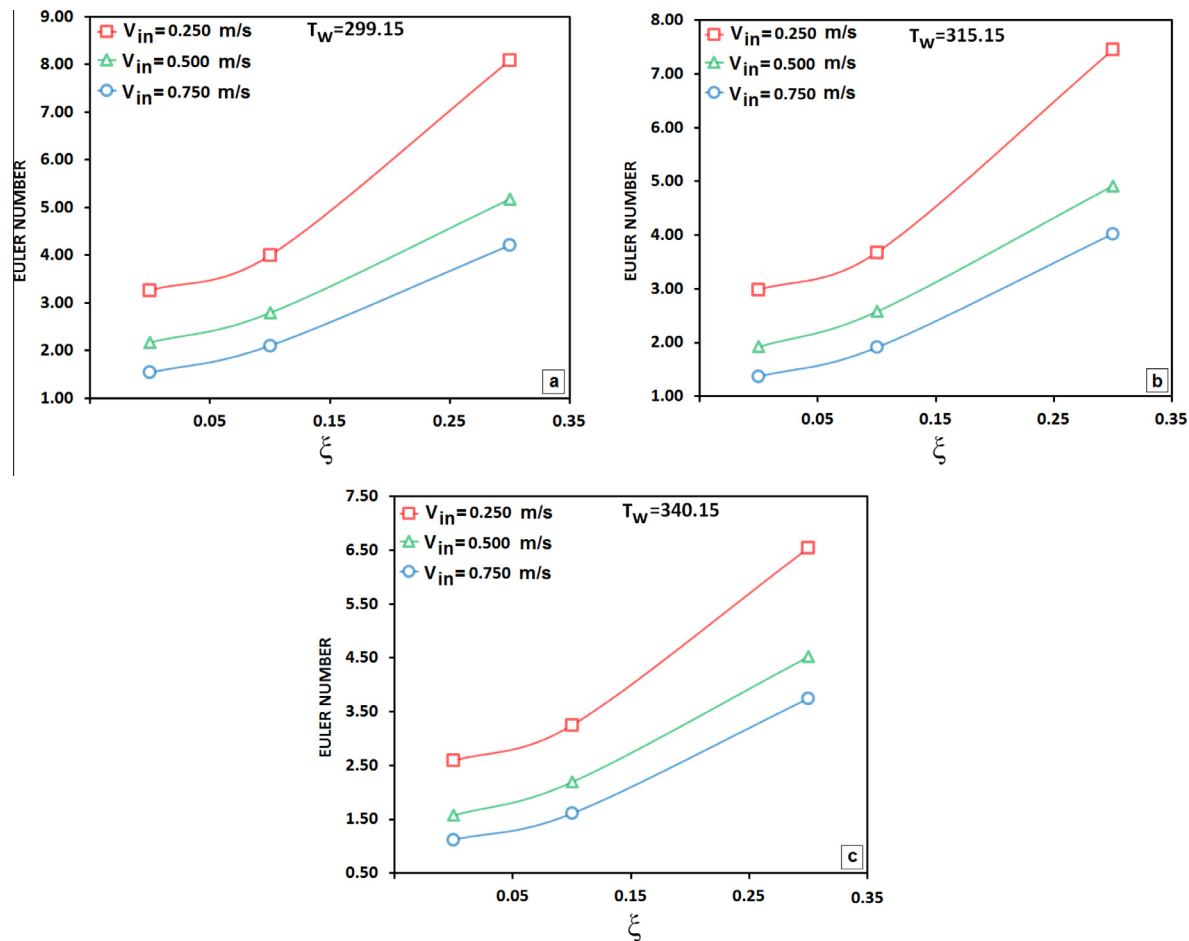


Fig. 6. Euler number variation versus NEPCM volume fraction at different inlet velocities for bottom wall temperatures of (a) 299.15, (b) 315.15, and (c) 340.15.

dimensionless parameter related to the ratio of pressure-drop of the system and squared Reynolds number, as defined in Eq. (12). Adding NEPCM particles to the base working fluid results in increasing this ratio because of the higher viscosity of the NEPCM slurries in comparison with the pure water. When the concentration of NEPCM particles in fluid increases, the fluid becomes more viscous and this dramatically increases the Euler number. Therefore, it can be realized that when the volume fraction of NEPCM particles increases, it enhances the cooling performance of the system, but on the other hand it affects pressure drop and increases the Euler number of the system, simultaneously.

Fig. 7 shows the contours of specific heat, viscosity, temperature and velocity at the surface passing the middle of the computational domain height. Since higher NEPCM volume fraction causes a higher viscosity as shown in Fig. 7(b), a higher pressure-gradient along the channel is expected. It is depicted in Fig. 7(b) and (c) that in regions close to the fin surface higher temperature results in lower viscosity. The reason is that the hotter molecules are enjoying higher energy level comparing with cold ones and these high energetic and less sluggish molecules, cause increase in kinetic energy of each molecule. Therefore, the attractive/repulsive intermolecular forces become more dominant that lead to a lower viscosity near the pins surfaces. In addition, when NEPCM volume fraction increases, stronger disturbance of the flow around the individual particles results in a higher internal shear stress in comparison with pure liquid. Increasing internal shear stress cause the fluid to be more viscous and sluggish. Since the NEPCM slurry, exhibits higher specific heat level between liquidus and solidus temperatures in comparison with the base fluid, as shown in

Fig. 7(a), a higher thermal energy is absorbed by them which leads to a lower temperature rise. Investigating Fig. 7(a), one can understand how the temperature rise of particles affects the specific heat of the slurry. NEPCM slurries are acting more efficiently compared with pure coolants. First, they have a higher heat capacity level when the temperature of them is somewhere between T_1 and T_2 . This lets the slurries to absorb more heat with lower temperature increase. Therefore, the slurries are ready to absorb more heat. Second, the higher heat capacity of the melting NEPCM resists the temperature increase and slows the temperature increase rate. This fact slows down the thermal boundary layer growth along the oblique fins surfaces. Therefore, it exhibits a higher temperature gradient, which boosts up the heat rejection rate. These two reasons improve the cooling performance of the system and convert the pure fluid coolant to a coolant of high quality. Observing the results presented in Fig. 7(d) and (c) reveals that the effect of the oblique channels cause the working fluid with a high specific heat value, flow from a channel to its neighbor channel and meanwhile absorb the heat from the hot fins. The role of the oblique fins to interrupt hydrodynamic and thermal boundary layers formation are worth paying attention. Velocity and temperature contours show how the higher NEPCM volume fraction leads to a thinner boundary layer compared to pure liquid and results in a higher heat dissipation rate. Using NEPCM slurry in a channel with oblique fins that are interrupting the boundary layer formation results in existence of longer and thinner boundary layers in comparison with conventional microchannels with pure coolants; this phenomenon enhances the cooling performance of the heat sink substantially.

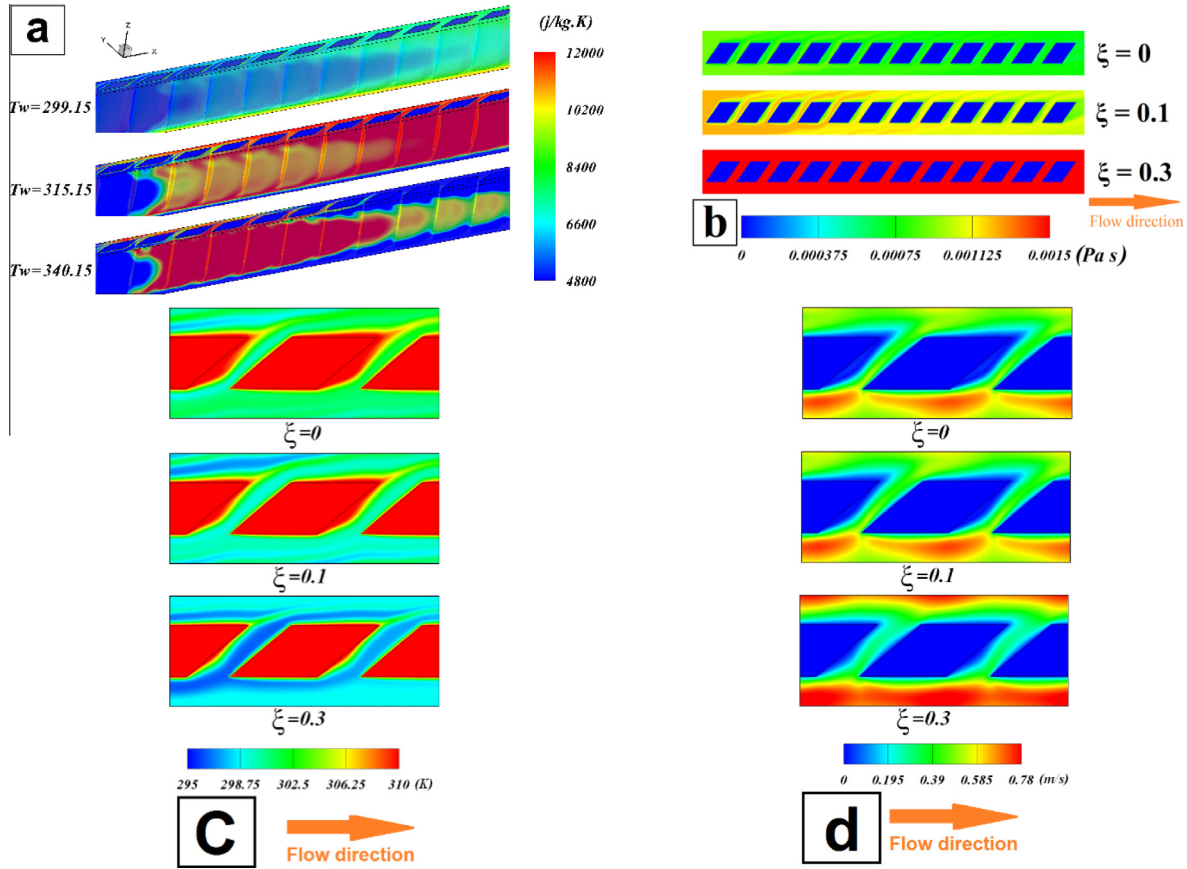


Fig. 7. The contours of (a) specific heat, (b) viscosity, (c) temperature and (d) velocity at the surface passing the middle of the computational domain height.

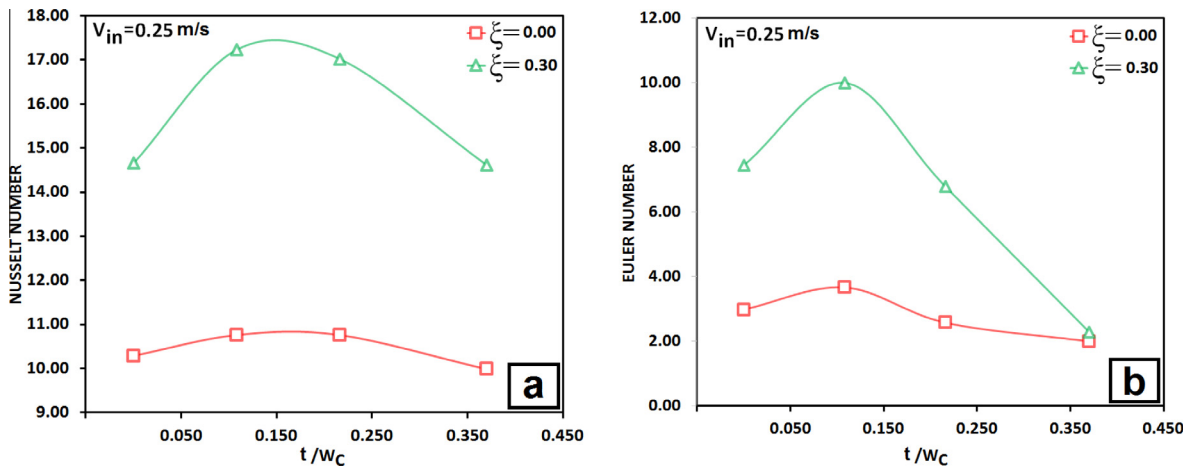


Fig. 8. Variations of Nusselt and Euler numbers versus tip clearance to channel width ratio at different NEPCM volume fractions.

6.2. Effect of tip clearance

As it was explained before, beside the desirable effect of presence of NEPCM particles in coolant on cooling performance of the system, a non-desirable higher pressure-drop is inevitable. Wang et al. [26] also reported a lower ratio of heat transfer to pumping power with higher Reynolds number in NEPCM slurries. The reason of this higher pressure drop is mainly the interactions between NEPCM particles and fluid flow. In this work, the thermal and hydrodynamics effects of tip clearance introduction over the fins on heat sink behavior under the condition of the fixed flow velocity

($V = 0.25$ m/s) are investigated. The simulated conditions have a bottom surface temperature of 315.15 K. NEPCM volume fraction is zero (pure water) or 0.3 and the tip clearance to channel width ratio ranges from 0.108 to 0.370.

As depicted in Figs. 8 and 9, for the heat sink with NEPCM slurry coolant, when tip clearance to channel width ratio (t/W_c) increases, the Nusselt and Euler numbers increase first and then decrease after they reach to their maximum values. It can be concluded from the illustrated results that heat sinks with NEPCM slurry coolant and with (t/W_c) ratios of 0.00–0.375, have a higher Nusselt number than the base zero-tip clearance condition. The maximum Nusselt

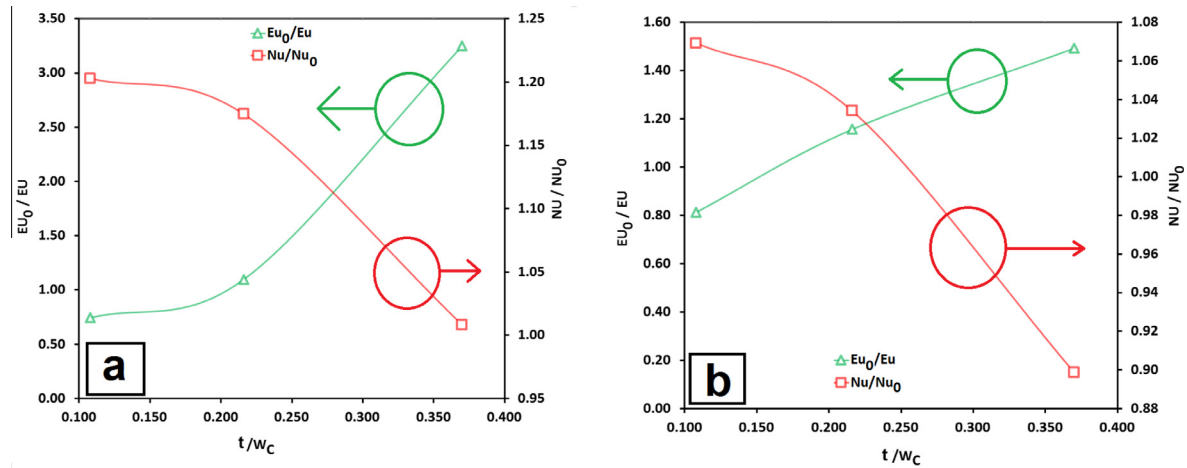


Fig. 9. Ratios of Nusselt and Euler numbers of the heat sink with tip clearance to the heat sink without tip clearance: (a) with NEPCM slurry coolant, and (b) pure water coolant.

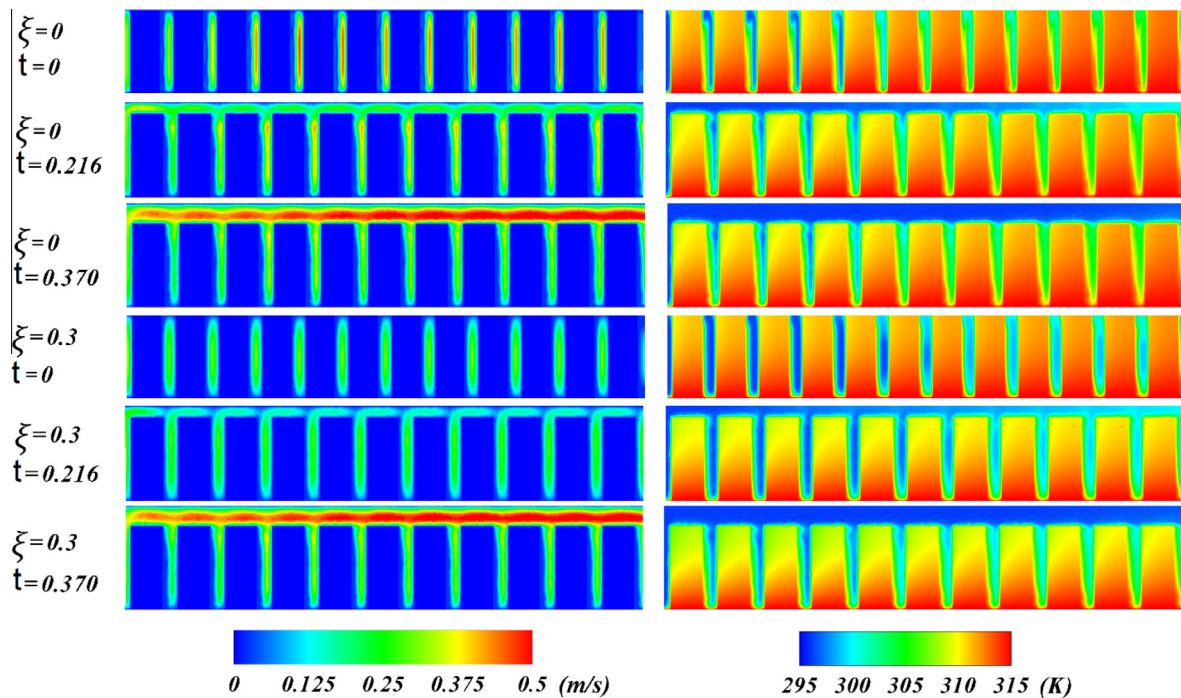


Fig. 10. Variation of (right side): temperature and (left side): velocity distribution in microchannel by introducing tip-clearance to the heat sink or employing NEPCM slurry coolant.

number for heat sinks with NEPCM slurry coolant is achieved when t/W_c ratio is about 0.15. On the other hand, Euler number of the heat sinks with NEPCM slurry coolant and with tip clearance, is above the base zero tip clearance condition when t/W_c is between 0.00 and 0.20. Therefore, if we consider the zero tip clearance heat sink as the base point, when $0.20 < t/W_c < 0.375$, the heat sink with NEPCM slurry coolant has a higher Nusselt number and lower Euler number in comparison with the base point. It is worth mentioning that the increase of the Nusselt number by increasing t/W_c ratio, in the heat sink with NEPCM slurry is stronger compared with heat sink with pure water. In Fig. 9, by defining the two parameters of Nu/Nu_0 and Eu_0/Eu , a comparison between thermal and hydrodynamic performance of the heat sink with tip clearance is compared to the heat sink with zero tip clearance. Both of these parameters are desired to be more than 1.00, so it means the higher Nusselt number and lower Euler number in comparison with the base

point. As it was illustrated in Fig. 9, the desired criteria for t/W_c - where both Nu/Nu_0 and Eu_0/Eu parameters value are above 1.00 - when using NEPCM slurry coolant is $0.20 < t/W_c < 0.375$ and when using pure water coolant is $0.16 < t/W_c < 0.26$.

The results in this study are an agreement with the results presented by Sparrow et al. [27]. Our data show that the suitable size of tip clearance leads to an enhanced cooling performance due to two reasons: (1) existence of the gap over the fins increases the total heat transfer area between hot fins surfaces and the cold coolant; and (2) if the tip clearance size, based on the results presented in Fig. 9, is adjusted appropriately, the top surface of the fins rejects more average heat flux in comparison with lateral surfaces and therefore, a substantially higher total Nusselt number would be achieved. The higher heat rejection rate from the top surface area of the fins is basically caused by induced three-dimensional effects via flow structures generated by the bypass flow and vortex

generation at the tips as compared to the reference quasi 1D channel flow without tip clearance. These three-dimensional effects would be vanished as t/W_c ratio increases. Interested readers are referred to [15] reported by Rozati et al. for further exploration of the effect of the tip clearance introduction on three dimensional flow generation over the fins. The trend of the results are in agreement with the results reported by Min et al. [14] for a conventional heat sink.

The contours depicted in Fig. 10 show how the velocity and temperature distributions in the microchannel are affected by introducing tip clearance to the heat sink or employing NEPCM slurry coolant. These contours are taken from the vertical surface passing from the midpoint of the bottom surface.

It can be observed in Fig. 10, that increasing the NEPCM volume fraction decreases the fins' temperature, which is due to thinner and longer boundary layer and higher temperature gradient in thermal boundary layers on lateral and top surfaces.

7. Conclusions

In this study, the thermal and hydrodynamic performance of a microchannel with oblique fins are enhanced by using NEPCM slurry coolant and introducing tip clearance over the fins. NEPCM slurry coolant could enhance the cooling performance of the heat sink substantially. The lower average bulk temperature and higher average specific heat of the coolant was observed when NEPCM slurry coolant is used. The disadvantage of utilizing NEPCM slurry as the coolant is higher Euler number, which needs a higher pumping power. This issue was addressed through introducing tip clearance to the microchannel. It was shown that, there is a range of t/W_c ratio in which the higher Nusselt number with a lower Euler number is expected. This desired range is $0.20 < t/W_c < 0.375$ when the microchannel is working with NEPCM slurry and $0.16 < t/W_c < 0.26$ when pure water employed as the coolant.

Conflict of interest statement

None declared.

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References

- [1] M.E. Steinke, S.G. Kandlikar, Single-phase liquid heat transfer in plain and enhanced microchannels, in: Proceedings of the 4th International Conference on Nanochannels, Microchannels, and Minichannels, Limerick, Ireland, 2006, pp. 943–951.
- [2] J.F. Tullius, T.K. Tullius, Y. Bayazitoglu, Optimization of short micro pin fins in minichannels, *Int. J. Heat Mass Transfer* 55 (2012) 3921–3932.
- [3] A. Kosar, Y. Peles, Thermal-hydraulic performance of MEMS-based pin fin heat sink, *ASME J. Heat Transfer* 128 (2) (2006) 121–131.
- [4] Y. Wang, F. Houshmand, D. Elcock, Y. Peles, Convective heat transfer and mixing enhancement in a microchannel with a pillar, *Int. J. Heat Mass Transfer* 62 (2013) 553–561.
- [5] Y. Wang, Y. Peles, An experimental study of passive and active heat transfer enhancement in microchannels, *ASME J. Heat Transfer* 136 (3) (2014) 031901, <http://dx.doi.org/10.1115/1.4025558>.
- [6] H.R. Seyf, Z. Zhou, H.B. Ma, Y. Zhang, Three dimensional numerical study of heat-transfer enhancement by nano-encapsulated phase change material slurry in microtube heat sinks with tangential impingement, *Int. J. Heat Mass Transfer* 56 (2013) 561–573.
- [7] S. Kuravi, K.M. Kota, J. Du, L.C. Chow, Numerical investigation of flow and heat transfer performance of nano-encapsulated phase change material slurry in microchannels, vol. 131, 2009, pp. 062901–1.
- [8] M. Goel, S.K. Roy, S. Sengupta, Laminar forced convection heat transfer in microencapsulated phase change material suspension, *Int. J. Heat Mass Transfer* 37 (4) (1994) 593–604.
- [9] R. Sabbah, J. Seyed-Yagoobi, S. Al-Hallaj, Heat transfer characteristics of liquid flow with micro-encapsulated phase change material: numerical study, *J. Heat Transfer* 133 (2011), <http://dx.doi.org/10.1115/1.4004450>.
- [10] Y.J. Lee, P.S. Lee, S.K. Chou, Enhanced thermal transport in microchannel using oblique fins, *J. Heat Transfer* 134 (2012). 101901–1.
- [11] Y.J. Lee, P.S. Lee, S.K. Chou, Numerical study of fluid flow and heat transfer in the enhanced microchannel with oblique fins, *J. Heat Transfer* 135 (2013). 041901–1.
- [12] M. Reyes, J.R. Arias, A. Velazquez, J.M. Vega, Experimental study of heat transfer and pressure drop in micro-channel based heat sinks with tip clearance, *Appl. Therm. Eng.* 31 (2011) 887–893.
- [13] K.A. Moores, Y.K. Joshi, Effect of tip clearance on the thermal and hydrodynamic performance of a shrouded pin fin array, *ASME J. Heat Transfer* 125 (2003) 999.
- [14] J.Y. Min, S.P. Jang, S.J. Kim, Effect of tip clearance on the cooling performance of a microchannel heat sink, *Int. J. Heat Mass Transfer* 47 (2004) 1099–1103.
- [15] A. Rozati, D.K. Tafti, Neal E. Blackwell, Effect of pin tip clearance on flow and heat transfer at low Reynolds numbers, *ASME J. Heat Transfer* 130 (2008). 071704–1.
- [16] Z. Liu, N. Ning, C. Zhang, Influence of tip clearance on heat transfer efficiency in micro-cylinders-group heat sink, *Exp. Therm. Fluid Sci.* 46 (2013) 64–73.
- [17] Y.W. Zhang, A. Faghri, Analysis of forced convection heat transfer in microcapsulated phase change material suspensions, *J. Thermophys. Heat Transfer* 9–4 (1995) 727–732.
- [18] A. Karnis, H.L. Goldsmith, S.G. Mason, The kinetics of flowing dispersions. I. Concentrated suspensions of rigid particles, *J. Colloid Interface Sci.* 22 (1966) 531–553.
- [19] R.W. Watkins, C.R. Robertson, A. Acrivos, Entrance region heat transfer in flowing suspensions, *Int. J. Heat Mass Transfer* 19 (1976) 693–695.
- [20] F.P. Incropera, D.P. DeWitt, Fundamentals of Heat and Mass Transfer, fifth ed., John Wiley & Sons, New York, 2002. Appendix Tables.
- [21] Y. Zhang, X. Hu, X. Wang, Theoretical analysis of convective heat transfer enhancement of microencapsulated phase change material slurries, *Heat Mass Transfer* 40 (2003) 59–66.
- [22] H. Schlichting, K. Gersten, Boundary Layer Theory, Springer, New York, 2000.
- [23] V. Vand, Viscosity of solutions and suspensions, *J. Phys. Colloid Chem.* 52 (1948) 300–321.
- [24] V. Vand, Theory of viscosity of concentrated suspensions, *Nature (London)* 155 (1945) 364–365.
- [25] E.L. Alisetti, S.K. Roy, Forced convection heat transfer to phase change material slurries in circular ducts, *J. Thermophys. Heat Transfer* 14 (2000) 115–118.
- [26] X. Wang, J. Niu, Y. Li, X. Wang, B. Chen, R. Zeng, Q. Song, Y. Zhang, Flow and heat transfer behaviors of phase change material slurries in a horizontal circular tube, *Int. J. Heat Mass Transfer* 50 (2007) 2480.
- [27] E.M. Sparrow, J.W. Ramsey, C.A.C. Altamiani, Experiments on in-line pin fin arrays and performance comparisons with staggered arrays, *J. Heat Transfer* 102 (1980) 44–50.